

# TPM Academy — Capstones

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The academy has **two capstone moments**: a mid-arc PRD discipline check in Week 3, and a full-arc artifact-set delivery in Week 8.

|                  | Week 3 — Mini-Capstone                                                  | Week 8 — Final Capstone                                                                |
|------------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Position         | End of the requirements arc                                             | End of the academy                                                                     |
| Subject          | <b>FieldPulse</b> (the shared HVAC dispatch SaaS case)                  | <b>Holocron</b> — an enterprise string-management problem, supplied as a problem brief |
| Duration         | 1 week (Mon–Fri)                                                        | 1 week (Mon–Thu build, Fri present)                                                    |
| AI tools         | <b>Off</b> — honor system                                               | <b>On</b> with validation discipline; new AI Spec Development pattern introduced Day 1 |
| Deliverable      | One Technical PRD + review-resolution log                               | 6-artifact set (PRD/TCD/TMD/SEP/DP + AI Spec)                                          |
| Compression rule | None — full 11-section PRD                                              | <b>2 pages per artifact</b> — discipline is what to cut                                |
| Quads            | Same quads since Week 1                                                 | Same quads since Week 1                                                                |
| Friday           | Cross-quad peer review (2 primary reviewers + 1 secondary, then revise) | 15-min quad presentations + 5 min Q&A; cohort + instructors score                      |
| Output proves    | Cohort can write a defensible PRD without AI                            | Muscles from Weeks 1–7 applied end-to-end on an unfamiliar enterprise problem          |

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## Week 3 — Mini-Capstone: PRD Discipline

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The first capstone is **focused and narrow**: produce one Technical PRD against the shared FieldPulse case, by hand, under peer review. AI is intentionally off all week so the **structured-thinking-expressed-as-written-words** muscle gets isolated and built.

### What ships Friday

- A revised PRD (template below)
- A **review-resolution log** showing which feedback was adopted, deferred, or rejected, with reasoning

### PRD template (sections used all week, cumulative)

1. Context
2. Problem
3. Goals & non-goals
4. Scope (in / out)
5. Solution sketch
6. Acceptance Criteria (Given/When/Then – granular, falsifiable)
7. Non-Functional Requirements (5 categories)
8. Metrics & validation (tied to Week 2 Tier Sheet)
9. Risks & open questions
10. Dependencies
11. Out-of-scope follow-ups

## Daily map

| Day | Topic                                                 | PRD section produced                                                           |
|-----|-------------------------------------------------------|--------------------------------------------------------------------------------|
| Mon | Drafting Technical PRDs                               | Sections 1–3 (context, problem, goals, scope)                                  |
| Tue | Granular Acceptance Criteria                          | 8–12 testable ACs                                                              |
| Wed | Non-Functional Requirements                           | NFR across performance / security / accessibility / observability / compliance |
| Thu | Mini-capstone PRD assembly                            | <b>Complete PRD</b> ready for review                                           |
| Fri | Peer review → revisions → secondary review → sign-off | <b>Revised PRD + resolution log</b>                                            |

## Review rubric (Friday)

| Dimension                                                  | Weight |
|------------------------------------------------------------|--------|
| Problem clarity                                            | 20%    |
| AC testability                                             | 25%    |
| NFR completeness                                           | 20%    |
| Strategy linkage (to Week 2 Tier Sheet / journey friction) | 15%    |
| Risk honesty                                               | 10%    |
| Writing discipline (no AI-generic prose)                   | 10%    |

## The non-AI rule

Allowed: spell-check, dictionary, customer-interview transcripts, Week 1–2 artifacts. Not allowed: any generative AI for drafting, summarizing, or critiquing any part of the PRD. The reasoning: AI fills vagueness with confident generic prose, and vagueness is the bug Week 3 is trying to eliminate.

**Full details:** [../../week-03/markdown/labs/README.md](#) .

Day-by-day lab briefs: [../../week-03/markdown/labs/day-](#)

[01..05-\\*.md](#) . Decks: [../../week-](#)

[03/markdown/presentations/day-01..05-\\*.md](#) .

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## Week 8 — Final Capstone: Integrated Artifact Set

The final capstone is **wide and integrative**. Every quad works the **Holocron** problem (brief issued end of Week 7) and produces a compressed-but-complete version of the 5-week artifact set, plus a new AI-generated integrated technical spec. Nothing is pre-specified — the brief gives business context and facts; the requirements are the quad's work.

### What ships Friday

```
capstone/
├── PRD-light.md      ← compressed from Week 3
muscle
├── TCD-light.md      ← compressed from Week 4
muscle
├── TMD-light.md      ← compressed from Week 5
muscle
├── SEP-light.md      ← compressed from Week 6
muscle
├── DP-light.md       ← compressed from Week 7
muscle
└── AI-Spec.md        ← NEW THIS WEEK – integrated
engineering-ready spec
```

**Compression constraint:** every artifact fits on **2 pages or less**. The discipline is what to cut, not what to add.

### Daily map

| Day | Topic                                       | Artifact produced                              |
|-----|---------------------------------------------|------------------------------------------------|
| Mon | AI Spec Development technique introduced    | AI Spec template + Holocron scope slice locked |
| Tue | Capstone discovery + compressed PRD         | PRD-light                                      |
| Wed | Capstone architecture + AI Spec drafted     | TCD-light + TMD-light + AI Spec v1             |
| Thu | Technical logic validation + finalization   | Validated, integrated capstone set             |
| Fri | <b>Final presentations</b> + course closure | Presented capstone; course complete            |

## Friday — presentation day

For ~6 quads: 15 min present + 5 min Q&A each, two morning blocks + two afternoon blocks, then cohort retrospective and post-assessments + course closure.

## Presentation rubric

| Dimension                                                          | Weight |
|--------------------------------------------------------------------|--------|
| Customer + problem clarity                                         | 15%    |
| Integrated artifact set (consistent cross-references)              | 15%    |
| <b>AI Spec quality</b> (engineering-ready; provenance log)         | 20%    |
| Trade-off honesty ( $\geq 2$ named with cost + revisit trigger)    | 15%    |
| Outcome thinking (NS / Tier-Sheet / leading-indicator vocabulary)  | 10%    |
| Delivery readiness (DP-light: backlog + tracking + one experiment) | 10%    |
| Presentation craft (all quad voices; time-managed; curious Q&A)    | 15%    |

## AI Spec Development (the new technique)

Day 1 introduces a structured prompt sequence that produces an engineering-ready integrated technical spec from PRD/TCD/TMD outputs, with explicit validation discipline. Day 3 applies it. The provenance log — prompts used, what was kept, what was discarded — is required and graded.

**Full details:** [../.../week-08/markdown/labs/README.md](#) .

Day-by-day lab briefs: [../.../week-08/markdown/labs/day-01..05-\\*.md](#) . Decks: [../.../week-08/markdown/presentations/day-01..05-\\*.md](#) .

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## How the two capstones relate

Week 3 proves the cohort can write **one well-formed PRD without AI**. Week 8 proves the cohort can produce the **full artifact set on an unfamiliar enterprise problem in five days, with AI used as a force multiplier instead of a crutch**. The Week 3 PRD muscle is the load-bearing input for the Week 8 PRD-light; every other compressed artifact rests on it.

Failing Week 3 makes Week 8 fragile. Passing Week 3 makes Week 8 an integration test rather than a relearning exercise.